SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks:50**

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I P.Sunayana(192424303) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 



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Substitution Vs Transposition: Which Provides Better Security?

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Introduction to Classical Encryption

- **Plaintext:** The original readable message.
- **Encryption:** The process of converting plaintext into ciphertext using an algorithm and, usually, a key.
- **Ciphertext:** The encrypted message that is difficult to understand without decryption.
- **Decryption:** The process of recovering the original message.
- **Example:**

HELLO

Encryption

KHOOR

- Caesar cipher with a shift of 3
- **Main goal:** Protect information from unauthorized access.

Substitution Cipher

Definition: A substitution cipher replaces each plaintext letter or symbol with another letter or symbol according to a fixed rule or key.

Working Principle:

- Each letter is replaced by another letter.
- The positions of the characters remain unchanged.
- Decryption reverses the substitution.

Advantages: Simple to implement and understand.

Limitation: Simple substitution ciphers can be broken using letter-frequency analysis and other cryptanalytic methods.

Example: Caesar Cipher (Shift = 3)

Plaintext	H	E	L	L	O
Ciphertext	K	H	O	O	R

Plaintext: HELLO

Ciphertext: KHOOR

Transposition Cipher

Definition: A transposition cipher rearranges the positions of plaintext characters without changing the characters themselves.

Working Principle:

- The message is arranged in a specific pattern.
- Characters are read in a different order.
- Decryption restores the original order using the correct key.

Advantages: Rearranges characters and can obscure their original positions.

Limitation: The original letters and their frequencies remain unchanged, so classical transposition is vulnerable to cryptanalysis.

Example: Rail Fence Cipher (2 Rails)

- **Plaintext:** HELLOWORLD

Rail 1: H L O O L

Rail 2: E L W R D

- **Ciphertext:** HLOOLELWRD

Key Differences Between the Two Techniques

Feature	Substitution	Transposition
Basic operation	Replaces characters	Rearranges characters
Character identity	Changes	Remains the same
Character order	Generally preserved	Changed
Letter frequencies	May change in simple cases	Preserved
Example	Caesar cipher	Rail Fence cipher

Security Analysis and Comparison

Substitution Cipher Security

- A Caesar cipher has only 25 nontrivial alphabet shifts. Monoalphabetic substitution preserves characteristic letter-frequency patterns.
- More complex polyalphabetic substitution can reduce some frequency patterns but remains vulnerable to modern cryptanalysis when used improperly.

Transposition Cipher Security

- The original character frequencies remain exactly the same. Short messages may reveal recognizable patterns.
- Security depends on the permutation key and the method used.

Which Provides Better Security?

Substitution vs Transposition

- A simple Caesar cipher is easy to break by testing possible shifts.
- A simple transposition cipher can be attacked by analyzing character patterns and possible rearrangements.
- A sufficiently complex transposition may be harder to break than a very weak substitution cipher, or vice versa.
- Combining substitution and transposition can provide stronger protection than either simple technique alone, but does not automatically guarantee modern security.

Final Conclusion: Neither classical substitution nor classical transposition guarantees better security in every case. For practical communication, authenticated encryption such as AES-GCM is the preferred choice.

THANK YOU